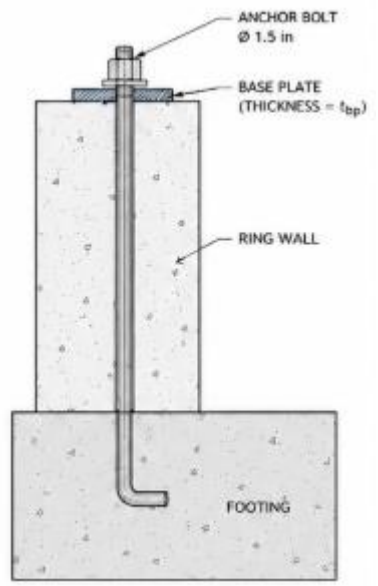
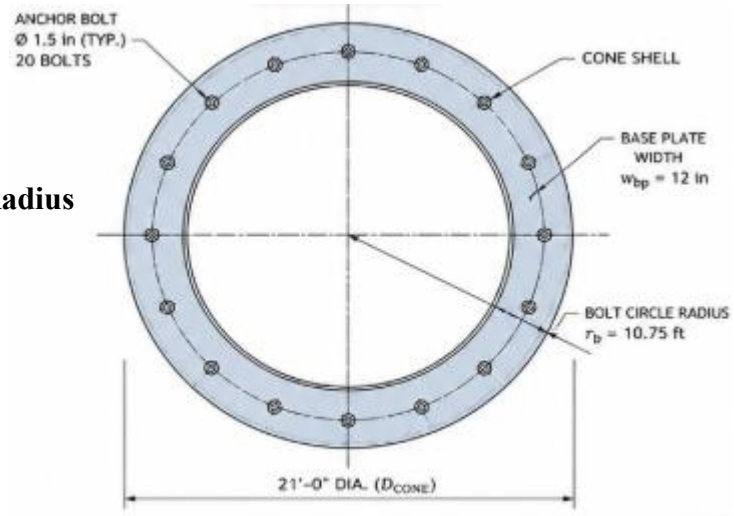
2. Base plate segment

• Parameter	Symbol	Value	Units
• Cone diameter	D_{cone}	25	ft
• Width of ring wall	Wrw	2	ft
• Number of anchor bolts	N_b	20	—
• Anchor bolt diameter	d_b	1.5	in
• Design Strip	N	12	in
• Vertical load on foundation	P	1650.2553	kips
• Governing overturning moment	M_u	2903.3789	kip-ft
• Steel yield strength	F_y	36	ksi
• compressive strength of concrete	F'_c	4	ksi
• Steel unit weight	γ_s	490	pcf
• Number of plate segments	N_s	4	—
• LRFD resistance factor	ϕ	0.90	—
• Modulus of elasticity	E	29000	ksi



1.1. Geometry Inputs Derived from These:

Outer Radius



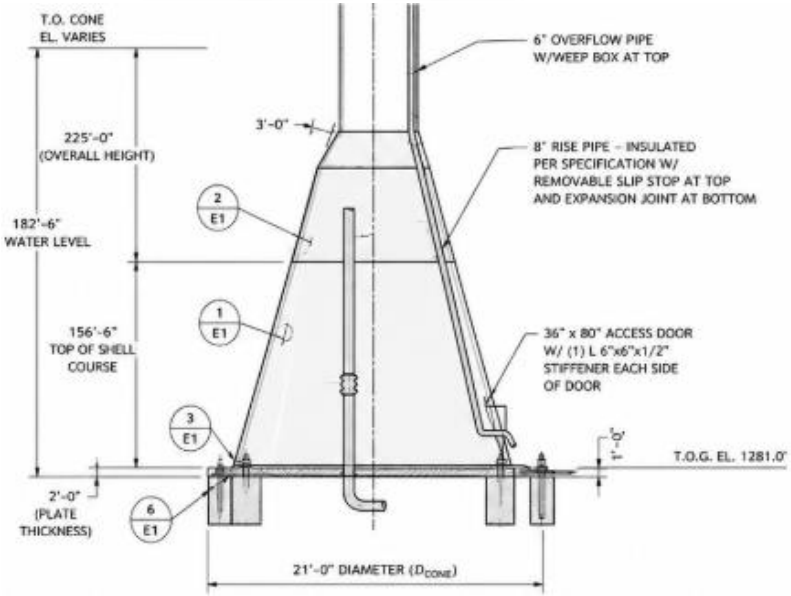
$$R_o = \frac{D_{cone}}{2} + \frac{w_{rw}}{4}$$

$$\boldsymbol{R_o = 11ft}$$

Inner Radius

$$R_i = \frac{D_{cone}}{2} - \frac{w_{rw}}{4}$$

$$\boldsymbol{R_i = 10ft}$$



2.0. Design Against Moment:

Based on the AISC LRFD moment design procedure adapted for annular strip behavior.

2.1. Determine Factored Load and Moment

Given

$P_u = 1650.2553$ kips

$M_u = 2903.4$ kip-ft

Convert moment to kip-in:

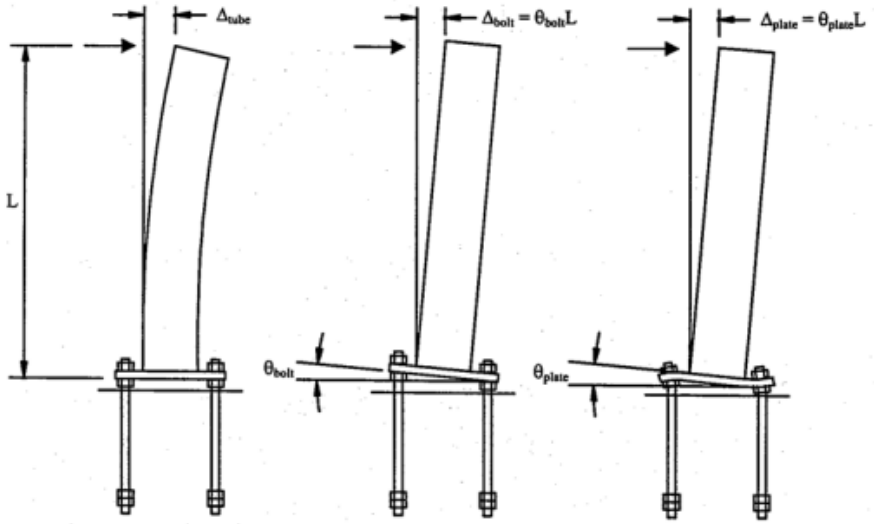
$$M_u = 2903.4(12) = \mathbf{34840.55\ kip\text{-}in}$$

2.2. Determine Circumferential Moment per Unit Strip

For annular strip design, distribute moment around the circumference.

Governing Equation

$$M_{strip} = \frac{M_u}{\pi D_{cone}}$$



Substitute:

$$M_{strip} = \frac{34840.55}{\pi(25)}$$

$$M_{strip} = \frac{34840.55}{78.5}$$

$M_{strip} = 36.97$ kip-in/ft

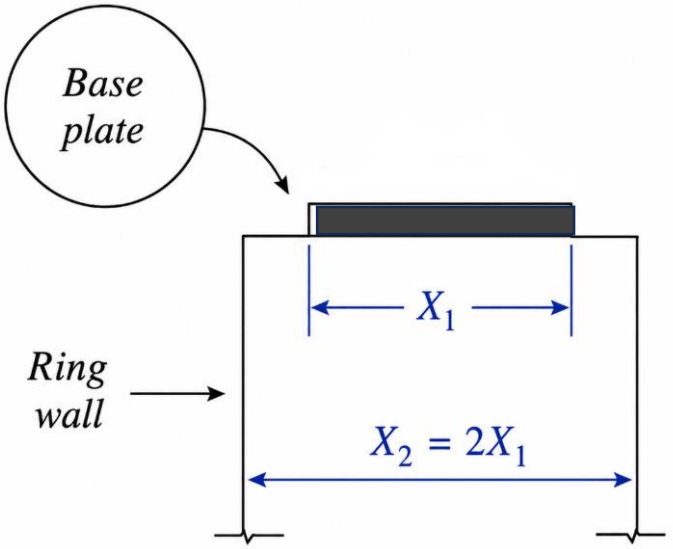
Convert to kip-in/in:

$$M_{strip} = \frac{36.97}{12}$$

$\boldsymbol{M_{strip} = 3.081}$ **kip-in/in**

2.3. Determine the maximum design bearing stress

$$\frac{A_2}{A_1} = \frac{x_2\,(rw)}{x_1\,(wbp)} = \frac{x_2}{x_1} = \frac{2}{1} = 2$$



$$F_p = 0.85\phi f_c' \sqrt{\frac{A_2}{A_1}} \leq 1.7 f_c'$$

$$0.85 * 0.9 * 4 * \sqrt{2} \leq 1.7 * 4$$

1.7×4=6.8

4.33≤6.8

X_1 = *width of ring wall*

X_2 = *width of base plate* = $2X_1$

CALCULATE F_p

if $F_p \leq 1.7 f_c'$ o.k.

If increase X_1 → maximum $\frac{X_2}{X_1} = 1.0$

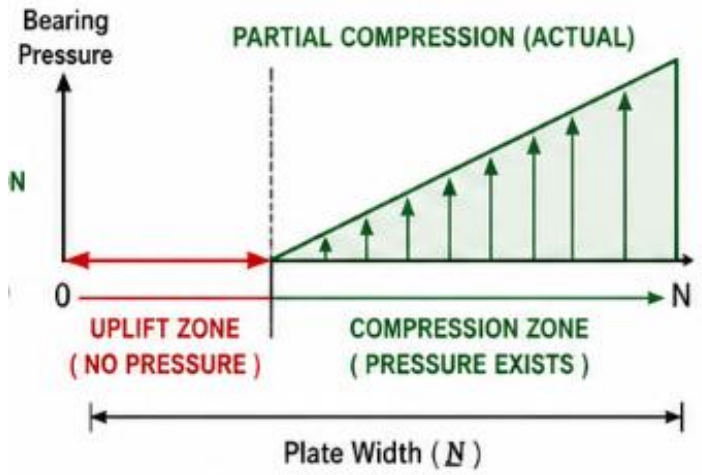
$$F_p \leq 1.7 f_c'$$

$$\frac{X_2}{X_1} = 1.0$$

3. Determine Equivalent Eccentricity

Governing Equation

$$e = \frac{M}{P}$$



Using strip quantities:

$$e = \frac{3.081}{2.084}$$

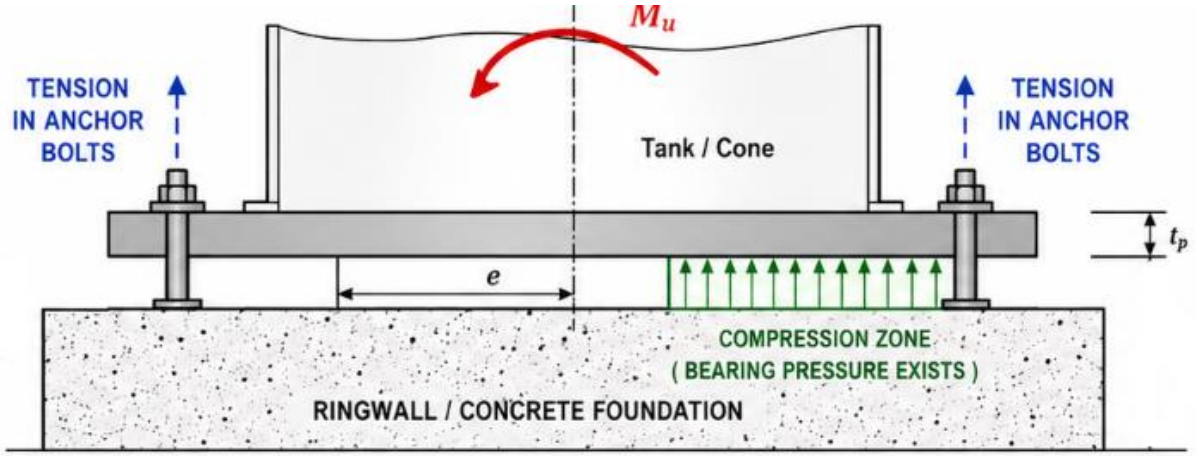
$$e = 0.62 \text{ in}$$

3.1. Check Bearing Condition

For full compression:

$e \leq \frac{N}{6}$

Substitute:



$$\frac{N}{6} = \frac{12}{6}$$

$\frac{N}{6} = 2$ in; Since: $\mathbf{2} > 0.62$

The are in tension

3.2. Determine Critical Section

Using annular strip geometry:

$$m = \frac{B}{2}$$

Given:

$$B = 12 \text{ in}$$

Thus:

$$m = \frac{12}{2}$$

$$m = 6 \text{ in}$$

$$m = \frac{12}{2} = 6 \text{ in}$$

3.3. Determine Plastic Moment per 1-in Strip

Using the AISC LRFD moment procedure:

Governing Equation

$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

3.4. Determine Strip Plastic Moment

Approximate factored strip moment:

$$M_{plu} = 73.48 \text{ kip-in/in}$$

3.5. Determine Required Thickness

Substitute:

$$t_p = \sqrt{\frac{4(3.081)}{0.90(36)}}$$

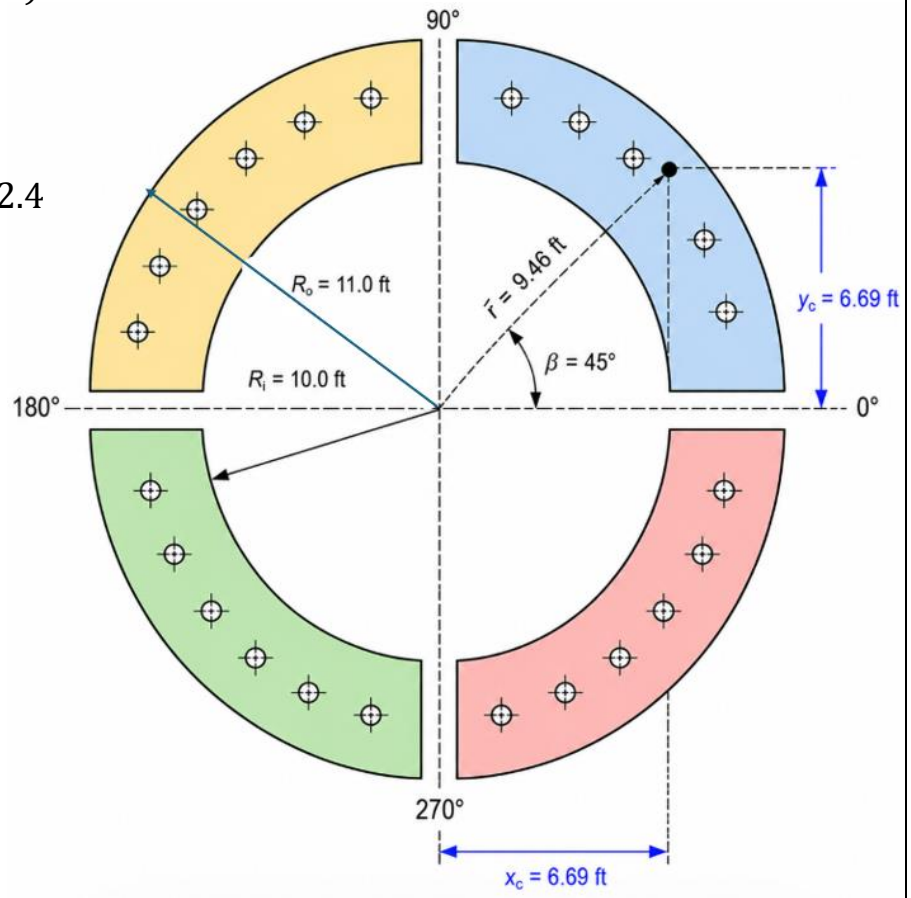
Denominator:

$$0.90(36) = 32.4$$

Numerator:

Thus:

$$t_p = 0.6 \text{ in}$$



3.6. Final Thickness Selection

From moment loading: $t_p = 0.6$ in

Therefore, moment governs. **Adopt** $\boldsymbol{t_p = 1.5}$ **in**

4.0. Local Uplift Bending Between Compression and Uplift Regions

the adopted thickness provides adequate resistance against:

- local uplift strip flexure,
- localized plate deformation,
- annular cantilever bending.

Therefore, local uplift bending is adequate.

5.0. Local Plate Stability / Buckling Check

The compactness ratio of the annular strip is:

$$\frac{B}{t_p} = \frac{12}{1.5} = 4$$

Because the plate is very stocky and compact:

- local plate buckling,
- compression instability,
- local flexural instability,

are not governing limit states.

Therefore, the annular plate is stable against local buckling.

6.0. Segment Angle

Since the annular base plate is divided into four equal segments:

$$\theta_{seg} = \frac{360^\circ}{4}$$

$$\theta_{seg} = 90^\circ$$

Convert to radians:

$$\theta_r = \frac{\pi}{2} = 1.5708 \text{ rad}$$

$$\theta_r = \frac{\pi}{2} = 1.5708 \text{ rad}$$

Plate Thickness

$$t_p = 1.5 \text{ in}$$

6.1. Centroid Radius of Annular Segment

Governing Equation

$$\bar{r} = \frac{4\sin(\theta_r/2)}{3\theta_r} \cdot \frac{R_o^3 - R_i^3}{R_o^2 - R_i^2}$$

Using:

$$R_o = 11.0 \text{ ft}$$

$$R_i = 10.0 \text{ ft}$$

$$\theta_r = 1.5708 \text{ rad}$$

Substituting:

$$\bar{r} = \frac{4\sin(1.5708/2)}{3(1.5708)} \cdot \frac{11^3 - 10^3}{11^2 - 10^2}$$

$$= \frac{4\sin(0.7854)}{4.7124} \cdot \frac{1331 - 1000}{121 - 100}$$

$$= \frac{4(0.7071)}{4.7124} \cdot \frac{331}{21}$$

$$= 0.6002(15.7619)$$

$$\boldsymbol{\bar{r} = 9.46 \text{ ft}}$$

6.2 Centroid Coordinates

For Segment 1:

$$\beta = 45^\circ$$

x-Coordinate

$$x_c = \bar{r}\cos\beta$$

Substitute:

$$x_c = 9.46\cos 45^\circ$$

$$x_c = 9.46(0.7071)$$

$$x_c = 6.69 \text{ ft}$$

$$x_c = 9.46\cos 45^\circ = 6.69 \text{ ft}$$

y-Coordinate

$$y_c = \bar{r}\sin\beta$$

Substitute:

$$y_c = 9.46\sin 45^\circ$$

$$y_c = 9.46(0.7071)$$

$$y_c = 6.69 \text{ ft}$$

$$y_c = 9.46 \sin 45^\circ = 6.69 \text{ ft}$$

6.3. Segment Centroid Coordinates

Segment	**Angular Range**	**Centroid Angle β**	**$x_c = \bar{r}\cos\beta$ (ft)**	**$y_c = \bar{r}\sin\beta$ (ft)**	**Centroid Coordinates (ft)**
Segment 1	$0^\circ \rightarrow 90^\circ$	45°	+6.69	+6.69	(6.69, 6.69)
Segment 2	$90^\circ \rightarrow 180^\circ$	135°	-6.69	+6.69	(-6.69, 6.69)
Segment 3	$180^\circ \rightarrow 270^\circ$	225°	-6.69	-6.69	(-6.69, -6.69)
Segment 4	$270^\circ \rightarrow 360^\circ$	315°	+6.69	-6.69	(6.69, -6.69)